

PHOTOSYNTHESIS IN HIGHER PLANTS

Comprehensive, verbatim-accurate, exam-oriented study notes preserving all terminology, equations, scientific contributions, physiological pathways (C_3 , C_4 , Photorespiration), and comparative matrices.

Target Exam: NEET / CBSE Class 11 / Biology Olympiad

Format: Multi-column Card Layout (Landscape)

Page Reference: NCERT Textbook Pages 133–152



CHAPTER 11 VISUAL MIND MAP

1. Fundamentals

- Autotrophic Nutrition
- Physico-chemical process
- Primary food source & O_2 release
- Chloroplast structure

2. Light Reaction

- Pigments: Chl a, b, Xanth, Carot
- LHC (PS I - P700, PS II - P680)
- Z-Scheme & Water Splitting
- Chemiosmotic Hypothesis

3. Dark Reaction

- Calvin C_3 Cycle (RuBP, PGA)
- Carboxylation, Reduction, Regen
- C_4 Pathway (Kranz Anatomy, PEP)
- Photorespiration (C_2 Cycle)

4. Factors & Revision

- Blackman's Law
- Light quality, intensity, duration
- CO_2 Conc. (C_3 vs C_4)
- Temp & Water Stress

11.1 What do we Know?

11.2 Early Experiments

11.3 Chloroplast Site

11.4 Pigment System

11.5 Light Reaction

11.6 Electron Transport

11.7 Calvin (C₃) Cycle

11.8 C₄ Pathway

11.9 Photorespiration

11.10 Rate Factors

NCERT Unit IV Overview Included

Melvin Calvin Nobel Spotlight

UNIT 4: PLANT PHYSIOLOGY OVERVIEW

TEXTBOOK PAGE 133

Biological study rests on two levels of organization:

1. Organismal & Above

Leads to Ecology, Evolutionary Biology, and related macro-disciplines.

2. Cellular & Molecular

Leads to Physiology and Biochemistry (e.g., Photosynthesis, Respiration, Growth).



SCIENTIST SPOTLIGHT: MELVIN CALVIN (1911–1997)

NOBEL PRIZE 1961

- **Education:** Ph.D. in Chemistry from Univ. of Minnesota; Professor at Univ. of California, Berkeley.
- **Discovery:** Post-WWII, Calvin & J.A. Bassham utilized radioactive **Carbon-14 (¹⁴C)** in algal photosynthesis studies to trace the pathway of carbon assimilation.
- **Core Principle:** Showed that plants convert light to chemical energy by transferring electrons in an organized pigment array. Mapped the complete biosynthetic path (Calvin Cycle).
- **Modern Application:** Principles are now utilized in solar energy research and renewable resource development.

11.1 WHAT DO WE KNOW? & 11.2 EARLY EXPERIMENTS

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11.1 CORE DEFINITION & CLASSICAL LEAF TESTS

Photosynthesis Definition: A *physico-chemical process* by which autotrophic green plants use light energy to drive the synthesis of organic compounds. It is the primary source of all food on earth and responsible for O₂ release.

1. Variegated / Black Paper Leaf Test:

Proves that **chlorophyll** and **light** are essential. Starch formation occurs exclusively in the green exposed regions.

2. Moll's Half-Leaf Experiment (KOH Test):

Enclosing half a leaf in a tube containing KOH-soaked cotton (which absorbs CO₂). Enclosed half tests negative for starch, exposed half tests positive. Proves **CO₂ is essential**.

11.2 CHRONOLOGICAL TIMELINE OF LANDMARK EXPERIMENTS

Joseph Priestley (1733–1804)

1770 / 1774

Performed bell jar experiments with a candle, mouse, and mint plant. Discovered **Oxygen (O₂)** in 1774.

Hypothesis: "Plants restore to the air whatever breathing animals and burning candles remove."

Jan Ingenhousz (1730–1799)

Aquatic Plant Test

Showed that **sunlight is essential** for air purification. Using aquatic plants, showed small O₂ bubbles formed around green parts only in bright light.

Key Finding: Only green parts release Oxygen in sunlight.

Julius von Sachs

1854

Provided evidence for **glucose production** during plant growth (stored as starch).
Showed green pigment (chlorophyll) is located in special cellular bodies (chloroplasts).

T.W. Engelmann (1843–1909)

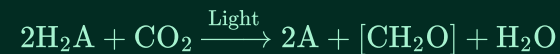
Action Spectrum

Split light using a prism to illuminate green alga *Cladophora* in aerobic bacteria suspension. Bacteria accumulated in **Blue and Red light** regions. Described 1st Action Spectrum.

Cornelius van Niel (1897–1985) — Milestone Contribution

Purple & Green Bacteria

Demonstrated that photosynthesis is essentially a light-dependent redox reaction in which hydrogen from a suitable oxidisable compound (H₂A) reduces carbon dioxide to carbohydrates:



- **In Green Plants:** H₂O acts as the hydrogen donor and is oxidised to Oxygen (O₂).
- **In Purple & Green Sulphur Bacteria:** H₂S acts as the hydrogen donor, yielding Sulphur or Sulphate (2A = 2S) instead of O₂.
- **Core Inference:** Oxygen (O₂) evolved by green plants originates exclusively from H₂O, NOT from CO₂ (experimentally verified later using heavy oxygen radioisotope ¹⁸O tracer techniques).

PHOTOSYNTHETIC CHEMICAL EQUATIONS COMPARISON

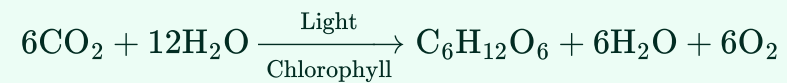
Historical evolution of the balanced chemical representation of photosynthesis from the 19th-century empirical formula to modern stoichiometry.

MID-19TH CENTURY EMPIRICAL EQUATION



$[\text{CH}_2\text{O}]$ represents a basic carbohydrate (e.g., glucose unit).

COMPLETE STOICHIOMETRIC NET EQUATION



Requires 12 H₂O molecules because 6 O₂ molecules (12 oxygen atoms) are evolved strictly from water splitting.

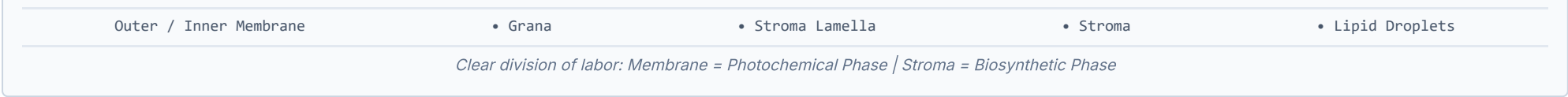
11.3 ORGANELLE ARCHITECTURE & DIVISION OF LABOUR

Mesophyll Cells

Photosynthesis occurs in green leaves and other green plant stems. Chloroplasts align along the walls of mesophyll cells to receive optimum incident light (flat surfaces parallel to walls under normal light, perpendicular under high light).

Chloroplast Region	Structural Components	Function & Reaction Type
Membrane System	Grana + Stroma Lamellae	Light Reactions (Photochemical): Traps light, splits water, synthesises ATP & NADPH.
Stroma Matrix	Enzymatic fluid matrix, Ribosomes, Starch granules, Lipid droplets	Dark Reactions (Carbon reactions): Enzymatic synthesis of sugars (CO_2 fixation). Dependent on products of light reaction (ATP/NADPH).

Figure 11.2 — Diagrammatic Section of Chloroplast



11.4 PAPER CHROMATOGRAPHY & LEAF PIGMENTS

4 Distinct Pigments

Separation of leaf pigments via **Paper Chromatography** reveals four distinct pigments:

1. Chlorophyll a

Color: Bright or Blue-green

Chief pigment (Reaction Centre)

2. Chlorophyll b

Color: Yellow-green

Accessory pigment

3. Xanthophylls

Color: Yellow

Accessory pigment

4. Carotenoids

Color: Yellow to Yellow-orange

Accessory pigment

Absorption vs. Action Spectrum (Figure 11.3 a, b, c)

- **Absorption Maxima:** Chlorophyll *a* shows peak light absorption in the **Blue and Red regions** of the visible spectrum (VIBGYOR).
- **Action Spectrum:** Maximum rate of photosynthesis (O_2 release) also occurs in Blue and Red regions, overlapping with Chl *a* absorption.

Role of Accessory Pigments (Chl b, Xanthophylls, Carotenoids):

- Absorb light at broader wavelengths and transfer energy to Chlorophyll *a*.
- Protect Chlorophyll *a* from **Photo-oxidation**.

11.5 LIGHT REACTION & 11.6 ELECTRON TRANSPORT (Z-SCHEME)

NCERT Pages 138–140

11.5 LIGHT HARVESTING COMPLEXES (LHC) & PHOTOSYSTEMS

The photochemical phase includes: light absorption, water splitting, oxygen release, and formation of high-energy chemical intermediates (**ATP and NADPH**).

Photosystems: PS I & PS II

Named in Order of Discovery

- Each photosystem consists of hundreds of pigment molecules bound to proteins forming an **Antennae system**.
- A single **Chlorophyll a molecule** forms the **Reaction Centre**.

Photosystem	Reaction Centre Peak	Primary Excitation Wavelength
Photosystem I (PS I)	P700	700 nm (Far Red Light)
Photosystem II (PS II)	P680	680 nm (Red Light)

11.6 ELECTRON TRANSPORT CHAIN & THE Z-SCHEME

Mechanism: PS II reaction centre (P680) absorbs 680 nm red light → electrons get excited and jump into an outer orbit → picked up by an **primary electron acceptor** → passed down an **Electron Transport System (cytochromes)** downhill on a redox potential scale to PS I.

Sequential Step-by-Step Z-Scheme Flow:

1. **PS II Excitation:** P680 absorbs 680 nm light → e^- transferred to primary acceptor.
2. **Downhill ETS:** e^- passed through Cytochrome complex to PS I (generates ATP).
3. **PS I Excitation:** P700 absorbs 700 nm light simultaneously → e^- excited to another acceptor with higher redox potential.
4. **NADP+ Reduction:** e^- moved downhill to NADP^+ → reduced to $\text{NADPH} + \text{H}^+$ by enzyme *NADP reductase*.

Why named "Z-Scheme"?

Formed when all carriers are placed in sequence on a redox potential scale (characteristic Z-shape pattern).

PS II → ETS → PS I → NADP⁺

11.6.1 PHOTOLYSIS / SPLITTING OF WATER

INNER THYLAKOID MEMBRANE

Electrons removed from PS II are continuously replaced by electrons supplied from the photolytic splitting of water.

WATER SPLITTING EQUATION



- **Location:** Water splitting complex is physically associated with **PS II**, located on the **inner side (lumen side)** of the thylakoid membrane.
- **Products:** Protons (H^+) and O_2 are released into the **Thylakoid Lumen**.

11.6.2 PHOTOPHOSPHORYLATION: NON-CYCLIC VS. CYCLIC

Phosphorylation: Process of ATP synthesis from $\text{ADP} + \text{P}_i$ in cells (mitochondria & chloroplasts) using light energy.

Feature	Non-Cyclic Photophosphorylation	Cyclic Photophosphorylation
Photosystems Involved	Both PS II and PS I operate in series	Only PS I is functional
Location	Grana Lamellae membranes	Stroma Lamellae (lacks PS II & NADP reductase)
Electron Flow	Non-cyclic Z-scheme flow	Electrons cycled back to PS I via ETS
Products Synthesised	Both ATP and NADPH + H^+	ATP only (No NADPH + H^+ formed)
Water Splitting / O_2	Occurs (O_2 evolved)	Absent (No O_2 evolved)
Dominant Condition	Normal photosynthetic conditions	Wavelengths beyond 680 nm available

11.6.3 CHEMIOSMOTIC HYPOTHESIS (PETER MITCHELL MECHANISM)

Proton Gradient across Thylakoid Membrane

ATP synthesis in chloroplasts is linked to the development of a **proton gradient** across the thylakoid membrane. Unlike mitochondria (where protons accumulate in the intermembrane space), in chloroplasts **protons accumulate inside the LUMEN**.

Step (a): Water Splitting

Water molecule splitting occurs on the *inner side of the membrane*. Protons (H^+) generated accumulate directly inside the **Lumen**.

Step (b): Proton Pump Carrier

Primary e^- acceptor transfers e^- to an H-carrier (Plastoquinone). It removes H^+ from the **Stroma** and releases H^+ into the **Lumen** when passing e^- to Cyt b_6f .

Step (c): NADP Reductase

NADP reductase enzyme faces the *Stroma side*. Reduction of $NADP^+$ to $NADPH + H^+$ consumes protons from the **Stroma**, further depleting stroma H^+ .

Breakdown of Proton Gradient & ATP Synthase Structure ($CF_0 - CF_1$)

- High H^+ in Lumen + Low H^+ in Stroma creates a measurable **pH drop in Lumen** & electrochemical gradient.
- Protons diffuse back to Stroma through the transmembrane channel of **CF_0 of ATP Synthase**.
- Breakdown of gradient provides energy to induce a conformational change in **CF_1 headpiece** (protrudes into stroma), catalyzing ATP synthesis.

4 Requirements for Chemiosmosis:

1. Thylakoid Membrane
2. Proton Pump
3. Proton Gradient ($pH_{lumen} < pH_{stroma}$)
4. ATP Synthase Enzyme (CF_0-CF_1)

11.7 DARK REACTION & CO₂ PRIMARY ACCEPTOR

Products of light reaction (ATP, NADPH, O₂): O₂ diffuses out; ATP & NADPH drive sugar synthesis in the **Biosynthetic Phase**.

Misnomer Alert: "Dark Reaction" does not mean it occurs in darkness; it is light-independent directly, but dependent on products of light reaction (ATP/NADPH). Stops shortly after light becomes unavailable.

Primary CO₂ Acceptor Discovery:

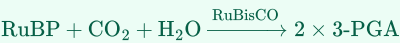
- Melvin Calvin used ¹⁴C in green algae and found the 1st stable product was a 3-carbon organic acid: **3-Phosphoglyceric Acid (3-PGA)**.
- Primary acceptor was unexpectedly proven to be a **5-carbon ketose sugar: Ribulose-1,5-bisphosphate (RuBP)**.

11.7.2 THE THREE STAGES OF THE CALVIN (C₃) CYCLE

The Calvin cycle operates in **ALL photosynthetic plants** (C₃, C₄, or CAM). It occurs in 3 main stages:

1 Carboxylation

Most crucial step. Fixation of CO₂ into RuBP catalysed by **RuBisCO**.



2 Reduction

Series of reactions forming glucose. Uses **2 ATP** (for phosphorylation) and **2 NADPH** (for reduction) per CO₂ fixed.

Requires 6 turns for 1 Glucose molecule.

3 Regeneration

Regeneration of the CO₂ acceptor RuBP to continue cycle uninterrupted.

Requires 1 ATP per CO₂ fixed.

Calvin Cycle Quantitative Stoichiometrics Ledger (Per 1 Glucose / 6 CO₂)

IN (Substrates Consumed)	OUT (Products Generated)	Energy Cost Breakdown per Glucose (C ₆ H ₁₂ O ₆)
6 CO ₂	1 Glucose (C ₆ H ₁₂ O ₆)	• Carboxylation + Reduction: 12 ATP + 12 NADPH
18 ATP	18 ADP + 18 P _i	• RuBP Regeneration: 6 ATP
12 NADPH	12 NADP ⁺	Total Cost: 18 ATP & 12 NADPH

11.8 C₄ PLANT SPECIAL ADAPTATIONS & KRANZ ANATOMY

Maize, Sorghum, Sugarcane

Plants adapted to dry tropical regions use the **C₄ Pathway (Hatch and Slack Pathway)**. First stable product is 4-carbon **Oxaloacetic Acid (OAA)**. They still use the C₃ Calvin cycle as the main sugar biosynthetic pathway!

Kranz Anatomy ('Kranz' = Wreath):

- Large **Bundle Sheath Cells** form several thick layers around vascular bundles.
- **3 Special Characteristics:** (1) Large number of chloroplasts, (2) Thick walls impervious to gas exchange, (3) No intercellular spaces.

Hatch & Slack Pathway Spatial Division of Labour:

Mesophyll Cells (Outer Layer)	Bundle Sheath Cells (Inner Layer)
• Primary CO ₂ Acceptor: Phosphoenolpyruvate (PEP) (3C).	• Contains RuBisCO enzyme in high abundance.
• Enzyme: PEP carboxylase (PEPcase) . Lacks RuBisCO!	• Lacks PEPcase enzyme .
• Fixes HCO ₃ ⁻ into C ₄ acid OAA → converted to Malic/Aspartic acid.	• C ₄ acid undergoes Decarboxylation to release CO ₂ and 3C acid.
• Receives 3C acid back from bundle sheath → regenerates PEP.	• Released CO ₂ enters Calvin C₃ Cycle to synthesise sugar!

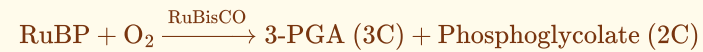
11.9 PHOTORESPIRATION (C₂ CYCLE / WASTEFUL PROCESS)

Present in C₃ / Absent in C₄

RuBisCO is the most abundant enzyme in the world. Active site can bind to both CO₂ and O₂ (competitive binding determined by relative concentration).

Mechanism in C₃ Plants:

When O₂ binds to RuBisCO, RuBP is converted into **1 molecule of 3-PGA (3C)** and **1 molecule of Phosphoglycolate (2C)** instead of 2 PGA molecules.



- NO ATP or NADPH synthesised.
- NO Sugar synthesised.
- Consumes ATP and releases CO₂ (Wasteful process!).

Why C₄ Plants Lack Photorespiration & Have Higher Yields:

C₄ plants have a **CO₂ Concentrating Mechanism**. Decarboxylation of C₄ acid in bundle sheath cells continuously pumps CO₂ directly to RuBisCO. High intracellular CO₂ ensures RuBisCO acts purely as a carboxylase, completely suppressing oxygenase activity!

FULL NCERT TABLE 11.1 SOLUTION — COMPARISON BETWEEN C₃ AND C₄ PLANTS

Completely filled comparative matrix based on NCERT options.

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Characteristics	C ₃ Plants	C ₄ Plants	NCERT Textbook Selection Options
Cell type of Calvin Cycle	Mesophyll cells	Bundle sheath cells	Mesophyll / Bundle sheath / Both
Cell type of initial Carboxylation	Mesophyll cells	Mesophyll cells	Mesophyll / Bundle sheath / Both
No. of cell types that fix CO ₂	One: Mesophyll	Two: Bundle sheath & Mesophyll	One / Two / Three
Primary CO ₂ Acceptor	RuBP (5-Carbon)	PEP (3-Carbon)	RuBP / PEP / PGA
No. of carbons in primary acceptor	5	3	5 / 4 / 3
Primary CO ₂ Fixation Product	PGA (3-Carbon)	OAA (4-Carbon)	PGA / OAA / RuBP
No. of carbons in primary product	3	4	3 / 4 / 5
Does the plant have RuBisCO?	Yes	Yes	Yes / No
Does the plant have PEPcase?	No	Yes	Yes / No
Which cells have RuBisCO?	Mesophyll cells	Bundle sheath cells	Mesophyll / Bundle sheath / None
CO ₂ fixation rate under high light	Medium / Low	High	Low / High / Medium
Photorespiration at low light	Negligible	Negligible / Absent	High / Low / Negligible
Photorespiration at high light	High	Negligible / Absent	High / Low / Negligible
Photorespiration at low CO ₂ conc.	High	Negligible / Absent	High / Low / Negligible
Photorespiration at high CO ₂ conc.	Negligible	Negligible / Absent	High / Low / Negligible
Optimum Temperature range	20°C – 25°C	30°C – 40°C	20-25°C / 30-40°C / Above 40°C
Examples	Wheat, Rice, Barley, Potato	Maize, Sorghum, Sugarcane	Common crop species

BLACKMAN'S (1905) LAW OF LIMITING FACTORS**CORE PRINCIPLE**

"If a chemical process is affected by more than one factor, then its rate will be determined by the factor which is nearest to its minimal value: it is the factor which directly affects the process if its quantity is changed."

Example: A green leaf in optimal light and CO₂ will not photosynthesise if temperature is near freezing. Adding heat (optimal temp) immediately restores rate.

1. LIGHT FACTOR

- Distinguish between light **quality**, **intensity**, and **duration**.
- Linear relationship at low light intensity.
- **Light Saturation:** Occurs at **10% of full sunlight!** Light is rarely limiting in nature (except shade/dense forest).
- High light beyond saturation breaks down chlorophyll (Photo-oxidation/destruction).

2. CO₂ CONCENTRATION

- **Major limiting factor** in nature (CO₂ is 0.03%–0.04% in atmosphere).
- Increase up to 0.05% enhances rate; beyond 0.05% is damaging long-term.
- **Saturation Levels:**
 - C₄ plants saturate at **360 μL L⁻¹**.
 - C₃ plants saturate at **450 μL L⁻¹**.
- **Greenhouse Crops:** Tomatoes & Bell Pepper grown in CO₂-enriched air yield higher.

3. TEMPERATURE FACTOR

- Dark reactions (enzymatic) are strongly temperature controlled. Light reactions are less affected.
- **C₄ plants:** Higher temperature optimum (30°C–40°C).
- **C₃ plants:** Lower temperature optimum (20°C–25°C).
- Tropical plants have higher optimum than temperate plants.

4. WATER FACTOR

- Indirect effect on plant rather than direct reactant effect.
- **Water Stress:** Causes stomata to close → reduces CO₂ availability.
- Causes leaves to **wilt** → reduces surface area and metabolic activity.



EASY WAY TO REMEMBER — EXAM MNEMONICS HUB

1. Leaf Chromatogram Pigment Colors

"B-Y-Y-O" (Bright-Yellow-Yellow-Orange)

- **Chl a:** Bright / Blue-green
- **Chl b:** Yellow-green
- **Xanthophyll:** Yellow
- **Carotenoid:** Yellow-Orange

2. Calvin Cycle 3 Stages

"C - R - R"

1. **Carboxylation** (RuBisCO attaches CO₂)
2. **Reduction** (Consumes ATP & NADPH)
3. **Regeneration** (Restores RuBP using ATP)

3. Chemiosmotic Requirements

"M - P - G - A"

- **M**embrane (Thylakoid)
- **P**ump (Proton Carrier)
- **G**radient (pH Difference)
- **A**TP Synthase (CF₀-CF₁)

Q1. By looking at a plant externally, can you tell whether a plant is C_3 or C_4 ? Why and how?

Answer: Generally, external morphology alone does not definitively distinguish C_3 and C_4 plants. However, plants adapted to dry, hot tropical environments with high light intensity (e.g., maize, sugarcane) are typically C_4 , whereas temperate crops (wheat, rice) are C_3 . Absolute confirmation requires internal anatomical study.

Q2. By looking at which internal structure of a plant can you tell whether a plant is C_3 or C_4 ? Explain.

Answer: By examining leaf cross-sections for **Kranz Anatomy** around vascular bundles. C_4 plants possess prominent concentric layers of large bundle sheath cells with thick walls, abundant chloroplasts, and no intercellular spaces. C_3 leaves lack distinct bundle sheath layers.

Q3. Even though very few cells in a C_4 plant carry out the Calvin cycle, why are they highly productive?

Answer: Because C_4 plants **lack photorespiration**. The Hatch-Slack mechanism concentrates CO_2 at the RuBisCO site in bundle sheath cells, ensuring RuBisCO operates strictly as a carboxylase. They also tolerate higher temperatures and utilize high light intensities efficiently.

Q4. RuBisCO acts as carboxylase and oxygenase. Why does it carry out more carboxylation in C_4 plants?

Answer: Decarboxylation of C_4 organic acid in bundle sheath cells continuously releases high concentrations of CO_2 locally around RuBisCO, ensuring $CO_2:O_2$ ratio remains high, suppressing oxygenation.



Key Locations & Enzymes

- **Water Splitting:** Lumen side of PS II
- **NADP Reductase:** Stroma side of membrane
- **C₃ Carboxylation:** Stroma (RuBisCO)
- **C₄ Initial Fixation:** Mesophyll (PEPcase)
- **C₄ Decarboxylation:** Bundle Sheath (RuBisCO)

Key Numerical Constants

- **Per Glucose in C₃:** 18 ATP + 12 NADPH
- **Per CO₂ in C₃:** 3 ATP + 2 NADPH
- **Light Saturation:** 10% of full sunlight
- **CO₂ Saturation:** C₄ = 360 $\mu\text{L L}^{-1}$ | C₃ = 450 $\mu\text{L L}^{-1}$
- **Atmospheric CO₂:** 0.03% – 0.04%

Key Scientific Names & Terms

- **Primary Acceptor C₃:** RuBP (5C Ketose)
- **Primary Product C₃:** 3-PGA (3C Acid)
- **Primary Product C₄:** OAA (4C Acid)
- **Alga in Engelmann Test:** *Cladophora*
- **Radioactive Tracer:** Carbon-14 (¹⁴C)